

Midterm Solutions:

1. (One More Observation; 25 pts)

Let $x_1, \dots, x_n, x_{n+1}$ denote some values of interest, and define the following descriptive statistics:

$$\bar{x}_n = \frac{1}{n} \sum_{i=1}^n x_i, \quad \sigma_n^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x}_n)^2.$$

Notice that these two statistics depend *only* on the first n numbers in the sample, and not on the final observation x_{n+1} .

Now, let $\bar{x}_{n+1}$ and σ_{n+1}^2 denote the sample mean and sample variance when the final observation x_{n+1} is taken into account:

$$\bar{x}_{n+1} = \frac{1}{n+1} \sum_{i=1}^{n+1} x_i, \quad \sigma_{n+1}^2 = \frac{1}{n+1} \sum_{i=1}^{n+1} (x_i - \bar{x}_{n+1})^2.$$

(a) Express $\bar{x}_{n+1}$ in terms of only n , $\bar{x}_n$ and x_{n+1} .

(b) Show that $\sum_{i=1}^n (x_i - \bar{x}_n) = 0$.

(c) Show that:

$$(n+1)\sigma_{n+1}^2 = n\sigma_n^2 + \frac{n}{n+1}(x_{n+1} - \bar{x}_n)^2.$$

[6pts] a) $\bar{x}_{n+1} = \frac{1}{n+1} \sum_{i=1}^{n+1} x_i$ by def.

$$= \frac{1}{n+1} \left(\underbrace{\sum_{i=1}^n x_i}_{= n \cdot \bar{x}_n \text{ by def'n}} + x_{n+1} \right)$$

$$\underline{\bar{x}_{n+1} = \frac{n}{n+1} \bar{x}_n + \frac{1}{n+1} x_{n+1}}$$

[6pts] b) $\sum_{i=1}^n (x_i - \bar{x}_n) = \left(\sum_{i=1}^n x_i \right) - \left(\underbrace{\sum_{i=1}^n \bar{x}_n}_{= n \cdot \bar{x}_n, \text{ since } \bar{x}_n \text{ doesn't depend on } i} \right)$

$$= \left(\sum_{i=1}^n x_i \right) - \underbrace{n \cdot \bar{x}_n}_{= \sum_{i=1}^n x_i \text{ by def'n}}$$

$$= \left(\sum_{i=1}^n x_i \right) - \left(\sum_{i=1}^n x_i \right)$$

$$\underline{\sum_{i=1}^n (x_i - \bar{x}_n) = 0}$$

[13pts] c) $(n+1)\sigma_{n+1}^2 = \sum_{i=1}^{n+1} (x_i - \bar{x}_{n+1})^2$ by def'n

$$= \left(\sum_{i=1}^n (x_i - \bar{x}_{n+1})^2 \right) + (x_{n+1} - \bar{x}_{n+1})^2$$

↓

$$= \left(\sum_{i=1}^n \left(\underbrace{x_i - \bar{x}_n}_{\text{Group these terms and}} + \underbrace{\bar{x}_n - \bar{x}_{n+1}}_{\text{expand}} \right)^2 \right) + (x_{n+1} - \bar{x}_{n+1})^2$$

$$= \sum_{i=1}^n \left[(x_i - \bar{x}_n)^2 + 2(x_i - \bar{x}_n)(\bar{x}_n - \bar{x}_{n+1}) + (\bar{x}_n - \bar{x}_{n+1})^2 \right] + (x_{n+1} - \bar{x}_{n+1})^2$$

$$= \underbrace{\left(\sum_{i=1}^n (x_i - \bar{x}_n)^2 \right)}_{= n \cdot \sigma_n^2} + 2 \cdot (\bar{x}_n - \bar{x}_{n+1}) \underbrace{\left(\sum_{i=1}^n (x_i - \bar{x}_n) \right)}_{= 0, \text{ by b)}} + (\bar{x}_n - \bar{x}_{n+1})^2 \underbrace{\left(\sum_{i=1}^n 1 \right)}_{= n} + (x_{n+1} - \bar{x}_{n+1})^2$$

(by distributing the sum, and pulling out constants)

$$= n \cdot \sigma_n^2 + n (\bar{x}_n - \bar{x}_{n+1})^2 + (x_{n+1} - \bar{x}_{n+1})^2$$

$$\begin{aligned} &= \left(\bar{x}_n - \left[\frac{n}{n+1} \bar{x}_n + \frac{1}{n+1} x_{n+1} \right] \right)^2, \text{ from a)} \\ &= \left(\frac{1}{n+1} [\bar{x}_n - x_{n+1}] \right)^2 \\ &= \frac{1}{(n+1)^2} (\bar{x}_n - x_{n+1})^2 \end{aligned}$$

$$= n \sigma_n^2 + \frac{n}{(n+1)^2} (\bar{x}_n - x_{n+1})^2 + (x_{n+1} - \bar{x}_{n+1})^2$$

$$\begin{aligned} &= (x_{n+1} - \left[\frac{n}{n+1} \bar{x}_n + \frac{1}{n+1} x_{n+1} \right])^2, \text{ by part a)} \\ &= \left(\frac{n}{n+1} [\bar{x}_n - x_{n+1}] \right)^2 \\ &= \frac{n^2}{(n+1)^2} (\bar{x}_n - x_{n+1})^2 \end{aligned}$$

$$= n \sigma_n^2 + \frac{n}{(n+1)^2} (\bar{x}_n - x_{n+1})^2 + \frac{n^2}{(n+1)^2} (\bar{x}_n - x_{n+1})^2$$

$$= n \sigma_n^2 + \frac{n^2 + n}{(n+1)^2} (\bar{x}_n - x_{n+1})^2$$

$$= \frac{n(n+1)}{(n+1)^2} = \frac{n}{n+1}$$

$$= n \cdot \sigma_n^2 + \frac{n}{n+1} (\bar{x}_n - x_{n+1})^2$$

2. (Discrete Random Variables; 25 pts)

Airlines often oversell tickets. Suppose that a plane with 100 seats, 104 passengers have tickets. Let X be the number of ticketed passengers who actually show up for the flight. Assume that the probability mass function of X , denoted $p(x)$, is given by the following:

x	96	97	98	99	100	101	102	103	104
$p(x)$	.15	.12	.14	.25	.17	.06	.05	.04	.02

If every passenger who does not get a seat needs to be compensated for \$300, whereas every ticket was sold for \$200. Let Z be the profit or loss of the airline due to overselling of 4 tickets, namely:

$$Z = \begin{cases} 4 * 200 - (X - 100) * 300 & \text{if } X > 100, \\ 800 & \text{otherwise.} \end{cases}$$

- Compute $\mathbb{P}(X \leq 100)$.
- What is the probability mass function of Z ?
- What is the expected profit, $\mathbb{E}[Z]$, and the variance of profit, $\text{Var}(Z)$?
- What is $\mathbb{E}[Z|X \leq 100]$ – the expected profit conditional on the airline **not** being over capacity?
(Hint: what is the conditional distribution of Z given that $X \leq 100$?)
- What is $\mathbb{E}[Z|X > 100]$?

[5 pts] a) $\mathbb{P}(X \leq 100) = \mathbb{P}(X=96) + \mathbb{P}(X=97) + \mathbb{P}(X=98) + \mathbb{P}(X=99) + \mathbb{P}(X=100)$

(since X can take no other value ≤ 100)

$= 0.15 + 0.12 + 0.14 + 0.25 + 0.17$ by the pmf. given

$\mathbb{P}(X \leq 100) = 0.83$

[5 pts] b) Since $Z = 800$ if $X \leq 100$, then $\mathbb{P}(Z=800) = \mathbb{P}(X \leq 100) = \underline{0.83}$ (from a)

Val. of X	$\mathbb{P}(X=x)$	Val. of Z
101	0.06	$800 - 1 \cdot 300 = 500$
102	0.05	$800 - 2 \cdot 300 = 200$
103	0.04	$800 - 3 \cdot 300 = -100$
104	0.02	$800 - 4 \cdot 300 = -400$

Putting these together:

z	800	500	200	-100	-400	otherwise
$P_Z(z)$	0.83	0.06	0.05	0.04	0.02	0

$$[5pts] c) E[Z] = \sum_z z \cdot P_Z(z)$$

$$= 800 \cdot 0.83 + 500 \cdot 0.06 + 200 \cdot 0.05 + (-100) \cdot 0.04 + (-400) \cdot 0.02$$

$$\underline{E[Z] = 692}$$

$$E[Z^2] = \sum_z z^2 \cdot P_Z(z)$$

$$= (800)^2 \cdot 0.83 + (500)^2 \cdot 0.06 + (200)^2 \cdot 0.05 + (-100)^2 \cdot 0.04 + (-400)^2 \cdot 0.02$$

$$E[Z^2] = 551,800$$

$$\text{So: } \text{Var}(Z) = E[Z^2] - (E[Z])^2$$

$$= 551,800 - 692^2$$

$$\underline{\text{Var}(Z) = 72,936}$$

[5pts] d) If $X \leq 100$, then $Z = 800$, no matter the specific value of X .

$$\text{Hence, } \underline{E[Z|X \leq 100] = E[800|X \leq 100] = 800}$$

[5pts] e) Many ways to show this.

Let $Y = 1\{X > 100\}$. Then:

$$\bullet IP(Y=0) = IP(X \leq 100) = 0.83$$

$$\bullet IP(Y=1) = IP(X > 100) = 0.17$$

Using the Tower Rule:

$$E[Z] = E[E[Z|Y]]$$

$$\underbrace{E[Z]}_{\substack{\text{"} \\ 692 \\ \text{(part c)}}} = \underbrace{E[Z|Y=0]}_{= 800 \text{ (part d)}} \cdot \underbrace{P(Y=0)}_{= 0.83} + \underbrace{E[Z|Y=1]}_{= 0.17}$$

$$692 = 800 \cdot 0.83 + E[Z|Y=1] \cdot 0.17$$

$$\Rightarrow \underline{E[Z|Y=1] = 164.7}$$

(Note: By construction, $E[Z|Y=1] = E[Z|X > 100]$)

3. (Standardizing a Normal / Gaussian Random Variable; 25 pts)

Recall that the cumulative density function of a Normal / Gaussian random variable with mean μ_X and variance σ_X^2 is given by:

$$F(x; \mu_X, \sigma_X^2) = \frac{1}{\sqrt{2\pi\sigma_X^2}} \int_{-\infty}^x e^{-\frac{(u-\mu_X)^2}{2\sigma_X^2}} du.$$

Let $X \sim N(\mu_X, \sigma_X^2)$ and define $Z = \frac{X - \mu_X}{\sigma_X}$.

- (a) For some real number $z \in \mathbb{R}$, express $\mathbb{P}(Z \leq z)$ in terms of the CDF of X .
- (b) Show that $\mathbb{P}(Z \leq z) = \Phi(z) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^z e^{-v^2/2} dv$, i.e. the CDF of Z is that of a standard normal random variable.

Note: Here you may **not** use the property from class stating that $\frac{X - \mu_X}{\sigma_X}$ has a standard normal distribution.

(Hint: substitute and perform a change of variables to simplify the term in the integral.)

$$\begin{aligned} [10 \text{ pts}] \text{ a) } \mathbb{P}(Z \leq z) &= \mathbb{P}\left(\frac{X - \mu_X}{\sigma_X} \leq z\right) \quad \text{by def'n of } Z \\ &= \mathbb{P}(X - \mu_X \leq z \cdot \sigma_X) = \mathbb{P}(X \leq \mu_X + z \cdot \sigma_X) \end{aligned}$$

$$\underline{\mathbb{P}(Z \leq z) = F_X(\mu_X + z \cdot \sigma_X)} \quad \text{by def'n of } F_X(\cdot)$$

[15pts] b) $IP(Z \leq z) = F_X(\mu_x + z\sigma_x)$ from a)

$$= \frac{1}{\sqrt{2\pi} \sigma_x} \int_{-\infty}^{\mu_x + z\sigma_x} \exp\left(-\frac{1}{2\sigma_x^2} (u - \mu_x)^2\right) du \quad \text{by def'n}$$

↓ A substitution: $u = \sigma_x \cdot v + \mu_x$
 $du = \sigma_x \cdot dv$

$$u = \mu_x + z\sigma_x \Leftrightarrow v = z$$

$$= \frac{1}{\sqrt{2\pi} \sigma_x} \int_{-\infty}^z \exp\left(-\frac{1}{2} \cdot \underbrace{\frac{1}{\sigma_x^2} (\sigma_x \cdot v + \mu_x - \mu_x)^2}_{= v^2}\right) \cdot \underbrace{\sigma_x}_{\text{pull out of integral}} dv$$

$$= \frac{1}{\sqrt{2\pi}} \cdot \cancel{\frac{1}{\sigma_x}} \cdot \cancel{\sigma_x} \cdot \int_{-\infty}^z \exp\left(-\frac{1}{2} v^2\right) dv$$

$IP(Z \leq z) = \Phi(z)$ as desired (by def. of $\Phi(\cdot)$)

4. (Joint Discrete Random Variables; 25 pts)

Consider the two discrete random variables X and Y with the following joint mass function.

$p(x, y)$		y		
		1	2	3
x	1	0	$\frac{1}{12}$	$\frac{1}{6}$
	2	$\frac{1}{12}$	$\frac{1}{3}$	$\frac{1}{12}$
	3	$\frac{1}{6}$	$\frac{1}{12}$	0

- Find the marginal probability mass functions of both X and Y .
- Are X and Y independent random variables? Justify your answer.
- Find $\text{Var}(X)$ and $\text{Var}(Y)$.
- Compute $\text{Cov}(X, Y)$.
- Find the probability mass function of the random variable $Z = X - Y$.
- What are $\mathbb{E}[Z]$ and $\text{Var}(Z)$?

[4pts] a) Using the joint pmf:

$p(x, y)$		y			x	$p_X(x)$
		1	2	3		
x	1	0	$\frac{1}{12}$	$\frac{1}{6}$	1	$\frac{1}{12} + \frac{1}{6} = \frac{1}{4}$
	2	$\frac{1}{12}$	$\frac{1}{3}$	$\frac{1}{12}$	2	$\frac{1}{12} + \frac{1}{3} + \frac{1}{12} = \frac{6}{12} = \frac{1}{2}$
	3	$\frac{1}{6}$	$\frac{1}{12}$	0	3	$\frac{1}{6} + \frac{1}{12} = \frac{1}{4}$

Summing up rows

$p_Y(y)$		1	2	3
		$\frac{1}{12} + \frac{1}{6} = \frac{1}{4}$	$\frac{1}{12} + \frac{1}{3} + \frac{1}{12} = \frac{6}{12} = \frac{1}{2}$	$\frac{1}{6} + \frac{1}{12} = \frac{1}{4}$

Summing up columns

[4pts] b) No. For instance, $p(1, 1) = 0$, but $p_X(1) \cdot p_Y(1) = \frac{1}{4} \cdot \frac{1}{4} = \frac{1}{16} \neq 0$

Hence, X and Y are not independent.

[4pts] c) Notice that X and Y have the same marginal pmf's (from a).

So, $\text{Var}(Y) = \text{Var}(X)$.

To compute: $\mathbb{E}[X] = \frac{1}{4} \cdot 1 + \frac{1}{2} \cdot 2 + \frac{1}{4} \cdot 3 = \underline{2}$

$$\mathbb{E}[X^2] = \frac{1}{4} \cdot 1^2 + \frac{1}{2} \cdot 2^2 + \frac{1}{4} \cdot 3^2 = \underline{4.5}$$

$$\text{Hence, } \text{Var}(X) = E[X^2] - (E[X])^2 \\ = 4.5 - 2^2$$

$$\Rightarrow \underline{\text{Var}(X) = 0.5}$$

By the argument above, $\underline{\text{Var}(Y) = \text{Var}(X) = 0.5}$ also.

[4pts] d) By defn, $\text{Cov}(X, Y) = \underbrace{E[X \cdot Y]}_{=2} - \underbrace{E[X]}_{=2} \cdot \underbrace{E[Y]}_{=2}$

$$= \sum_{x,y} x \cdot y \cdot p(x,y)$$

$$= \cancel{(1 \cdot 1) \cdot 0} + (1 \cdot 2) \cdot \frac{1}{12} + (1 \cdot 3) \cdot \frac{1}{6} + (2 \cdot 1) \cdot \frac{1}{12} + (2 \cdot 2) \cdot \frac{1}{3} \\ + (2 \cdot 3) \cdot \frac{1}{12} + (3 \cdot 1) \cdot \frac{1}{6} + (3 \cdot 2) \cdot \frac{1}{12} + \cancel{(3 \cdot 3) \cdot 0}$$

$$= \frac{11}{3}$$

$$\text{Cov}(X, Y) = \frac{11}{3} - 2 \cdot 2$$

$$\Rightarrow \underline{\text{Cov}(X, Y) = -\frac{1}{3}}$$

[4pts] e) $IP(Z = -2) = IP(X = 1, Y = 3) = p(1, 3) = \frac{1}{6}$

$$IP(Z = -1) = IP(X = 1, Y = 2) + IP(X = 2, Y = 3) \\ = p(1, 2) + p(2, 3) \\ = \frac{1}{12} + \frac{1}{12} = \frac{1}{6}$$

$$IP(Z = 0) = IP(X = 1, Y = 1) + IP(X = 2, Y = 2) + IP(X = 3, Y = 3) \\ = p(1, 1) + p(2, 2) + p(3, 3) \\ = 0 + \frac{1}{3} + 0 = \frac{1}{3}$$

By symmetry of X and Y (i.e. $f(x, y) = f(y, x)$), we can see that

$$IP(Z = 1) = IP(Z = -1) = \frac{1}{6}$$

$$IP(Z = 2) = IP(Z = -2) = \frac{1}{6}$$

All together:

z	-2	-1	0	+1	+2
$P_Z(z)$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{3}$	$\frac{1}{6}$	$\frac{1}{6}$

[5pts] f) By linearity of Expectation, $E[Z] = E[X - Y] = E[X] - E[Y]$

Since $E[X] = E[Y] = 2$, then: $E[Z] = 2 - 2 = 0$

Hence, $Var(Z) = E[Z^2] - (0)^2$

$$= \frac{1}{6} \cdot (-2)^2 + \frac{1}{6} \cdot (-1)^2 + \frac{1}{3} \cdot (0)^2 + \frac{1}{6} \cdot (1)^2 + \frac{1}{6} \cdot (2)^2$$

$$\underline{\underline{Var(Z) = 5/3}}$$
